



Seminar Paper

Predictive Modelling For Wheel Failure Estimation Within The Next 30 Days

Failure prediction for wheels based on sensor data and
railcar attributes

Michael Mutahi Kiboi

Born on: 11th July 1988 in Kenya

Matriculation number: 5200245

Shih-Hsien Hsu

Born on: 28th June 2000 in Taiwan

Matriculation number: 5261198

Wenjing Yu

Born on: 1st September 2000 in China

Matriculation number: 5268750

29th August 2025

Supervising professor

Prof. Dr. rer. pol. Pascal Kerschke

Contents

1	Introduction	1
1.1	Background of the Project	1
2	Data and Preprocessing	2
2.1	Dataset Overview	2
2.2	Data Cleaning	2
2.3	Data Merging and Sampling	3
2.3.1	Data sampling	3
2.3.2	Data Merging	3
2.4	Exploratory Data Analysis	4
3	Modeling and evaluation	6
3.1	Feature Engineering and Selection	6
3.1.1	Feature Engineering	6
3.1.2	Feature selection	7
3.2	Benchmarking and Model Selection	8
3.2.1	Benchmarking full model	9
3.2.2	Principal Component Analysis	11
3.2.3	Benchmarking Model on Reduced Features	11
3.2.4	Model Selection	12
3.3	Hyperparamter Optimization	12
3.4	Interpretability	13
3.4.1	Feature Importance (Cross-Entropy Loss)	13
3.4.2	Local Explanation (LIME-style)	14
3.4.3	Shapley Value Analysis (SHAP)	14
4	Conclusion	15

List of Figures

2.1	Count of failure reason	4
2.2	Mileage distribution by failure reasons	5
2.3	Boxplot of max vertical by failure	5
2.4	Distribution of equipment age(days)	5
3.1	wrappers vs importance filter	8
3.2	Task with failure reason as target	9
3.3	Full model Benchmark	10
3.4	PCA vs Benchmark models	11
3.5	Benchmark on reduced features	11
3.6	Feature importance by cross-entropy loss	13
3.7	Result of Local Explanation	14
3.8	Result of Game-theoretic fairness	14

List of Tables

2.1	Overview of RAS Detector Datasets	2
2.2	Feature Capping Ranges for WPD Dataset	3
2.3	Common features across different datasets data_merge.	4
3.1	Selected Features	8
3.2	Multiclass Log Loss Scores for Different Learners	10
3.3	Test set log-loss results for trained models.	11
3.4	Hyperparameter search space for XGBoost	12
3.5	Predictive Performance of XGBoost model	12
3.6	Confusion matrix of the tuned XGBoost model on the test set.	13
4.1	Contributions of the group members	16

List of Abbreviations

WPD	Wheel Profile Detector
THD	Truck Hunting Detectors
WILD	Wheel Impact Load Detectors
CV	Cross Validation
PCA	Principal Component Analysis
SVM	Support Vector Machines

1 Introduction

1.1 Background of the Project

Wheelsets, which consist of two wheels connected by an axle, are essential components of railway vehicles. Their condition has a direct impact on both the safety and efficiency of train operations. If a wheel develops defects such as cracks, flat spots, or excessive flange wear, it can damage the track, shorten the lifespan of equipment, and even increase the risk of derailments. For this reason, continuous monitoring and predictive maintenance is needed.

There are several types of detectors used to track wheel conditions. Wheel Impact Load Detectors (WILD) measure the vertical forces between the wheel and the rail, detecting defects such as flat spots. Truck Hunting Detectors (THD) monitor the side-to-side oscillations of trucks, which can reveal instabilities that accelerate wear. Wheel Profile Detectors (WPD) collect detailed measurements of wheels, such as flange height, rim thickness, which are used to decide whether a wheel meets the criteria for replacement. Together with mileage and failure records, these measurements provide valuable data for predictive maintenance.

In the United States, the Federal Railroad Administration (FRA) and the Association of American Railroads (AAR) have established rules to define “condemnable failures.” These rules specify when a wheel must be replaced to ensure safety. For modeling, failure reasons are concluded into four types: High Flange, High Impact, Thin Flange and Other.

The 2025 Railway Application Section Problem Solving Competition builds on these information with the target to create models that predict wheel failures within 30 days. The dataset covers seven years and includes railcar information, mileage records, wheel profile data, and detector records from WPD, THD, and WILD. Log-loss is chosen as the main evaluation metric.

This seminar paper presents a predictive modeling work for wheel failure prediction. It applies data preprocessing, feature engineering, model benchmarking, and interpretability. Different machine learning methods are evaluated, including Random Forest, XGBoost, Logistic Regression, and Support Vector Machines, with benchmarking carried out in the mlr3 framework. It also deals with data issues like missing values, imbalanced classes, and high-dimensional features. In addition, interpretability methods are used to provide insights into the key drivers of wheel failures.

2 Data and Preprocessing

2.1 Dataset Overview

The main data set comprises sensor data set

Dataset 1 (Main): Wheel sensor measurements (columns: traindate, equipmentnumber, truck, axle, side).

Dataset 2 (Failures): Failure labels (recordmonth, equipmentnumber, truck, axle, side, failedin30days).

Dataset 3 (Mileage): Aggregated mileage stats (recordmonth, equipmentnumber).

Dataset 4 (Alerts): Hunting index and load alerts (traindate, equipmentnumber, truck).

Dataset 5 (Vertical Dynamics): Stress-related metrics by axle and side.

Table 2.1: Overview of RAS Detector Datasets

Dataset	Features	Observations
Wheel Profile Detector (WPD)	18	4.57 million
Truck Hunting Detectors (THD)	12	2.66 million
Wheel Impact Load Detectors (WILD)	12	20.4 million
Mileage	6	101,000

2.2 Data Cleaning

Before proceeding with further analysis, data cleaning is necessary, with the primary task being the elimination of missing values and outliers. Load essential datasets and check the missing values. Select the suitable method to handle the missing data.

- The failure masked data has 440 missing values, and all of them pertain to the unlabeled truck type. All missing values are dropped directly since there is no method to view the correct truck type.
- WPD data cleaning: In wheel profile detector data, the technical standard is adapted to filter the outliers. No more missing values are shown after this step. Table 2.2 displays the technical standards of different parts of the train and rail(following the AAR wheel standards and FRA inspection guidelines).

Table 2.2: Feature Capping Ranges for WPD Dataset

Feature	Lower Bound	Upper Bound
Diameter of the wheel	31.0	43.0
Width of the wheel	5.3	5.9
Height of the flange	1.0	1.4
Thickness of the flange	0.75	1.4
Slope of the flange	0.25	0.6
Hollow of the tread	0.00	0.08
Thickness of the rim	1.0	2.0
Gauge from back to back	52.0	54.0

- THD data has 5418 missing values, which occupy only 0.2%, and can be dropped directly.
- WILD data has 300 missing values, which occupy less than 0.01%, and can be dropped directly.
- A total of 21,244 missing values were found across loaded and empty travel mileage, with each feature contributing 10,622 missing entries. These rows represent trains that did not move, as confirmed by unchanged total mileage compared to last month. Both features were imputed with zero to reflect a stationary status and preserve semantic integrity.

2.3 Data Merging and Sampling

After the data was cleaned and prepared for the model, it was required to merge all the different datasets. For saving the compute resources, we sampled 10,000 from failure masked data.

2.3.1 Data sampling

To verify that the sample reflects the whole dataset, we performed two goodness-of-fit tests. A Kolmogorov–Smirnov test showed no significant difference in the distribution of part mileage ($D = 0.0097$, $p \approx 0.3126$). For the binary variable “failed in 30 days,” both chi-squared and Fisher’s exact tests confirmed identical proportions ($p \approx 1$). These results indicate that the sample faithfully represents the population.

2.3.2 Data Merging

- Extracted unique equipment IDs and record months from the sampled failure data to define the scope of analysis.
- Standardized date formats across all datasets by converting relevant date fields to "YYYY-MM" strings, enabling consistent monthly joins.
- Merged the sampled failure data with the common features across the separate datasets.

Dataset	record month	equipment number	truck	axle	side
failure sample	✓	✓	✓	✓	✓
mileage	✓	✓			
THD	✓	✓	✓		
WILD	✓	✓	✓	✓	✓
WPD	✓	✓	✓	✓	✓

Table 2.3: Common features across different datasets data_merge.

- Check the missing value then removed redundant columns such as 'equipment initial', which was constant across sources and did not contribute additional information.
- Verified consistency across multiple 'train date' columns, retained only matching records, and unified the date field to a single 'train date' column. This is the vital step to eliminate the inflation.

2.4 Exploratory Data Analysis

Exploratory Data Analysis (EDA) has been performed to understand the patterns in wheel failures. The main target variable indicates whether a wheel failed within 30 days. As shown in Figure 2.1, the dataset is highly imbalanced: the majority of wheels did not fail (27496), while only a small proportion are labeled as failures. Among the failure cases, high impact (247) accounts for the most, high flange (40) and thin flange (51) second, and other (9) accounts for the least.

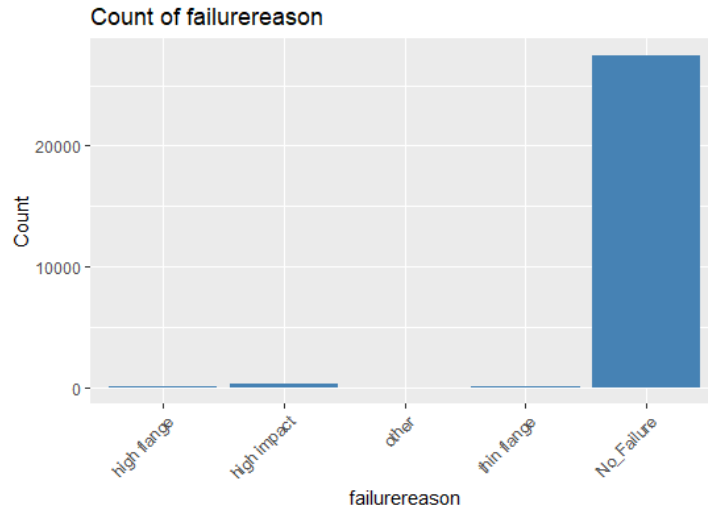


Figure 2.1: Count of failure reason

While failure counts are dominated by high impact, the mileage distribution shows a different outcome. Figure 2.2 presents the mileage distribution across different failure reasons. It can be observed that wheels failure of high flange and other tend to have higher mileage compared to those failure due to high impact or thin flange. The non-failure group ("No Failure") shows the widest mileage distribution, with several extreme outliers above 80,000 miles. This suggests that mileage plays an important role in wheel failure, but the exact failure mechanism varies by type.

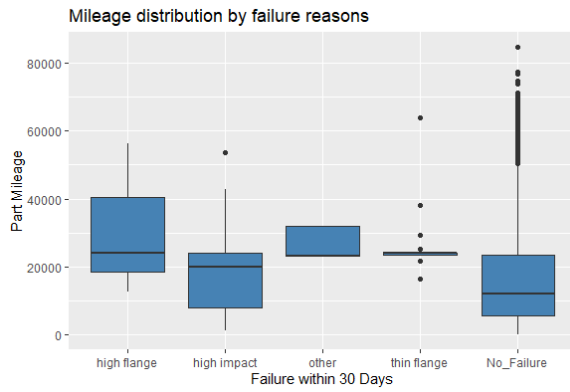


Figure 2.2: Mileage distribution by failure reasons

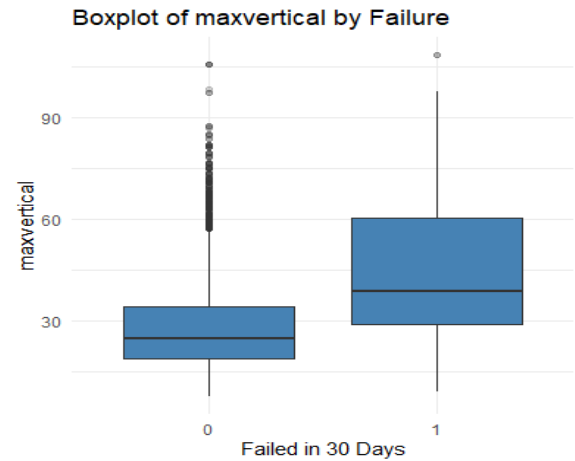


Figure 2.3: Boxplot of max vertical by failure

Figure 2.3 shows the differences between failed(=1) and non-failed(=0) wheels. The boxplots show that although most failures occur at moderate mileage levels, wheels that are still in use without failure can accumulate much higher mileage. This highlights the heterogeneous nature of wheel wear: some wheels can tolerate long using distances, while others fail earlier because of wear.

Figure 2.4 shows the distribution of equipment age measured in days. The majority of equipment is relatively new, below 1500 days. Only a small number of railcars are older than 3000 days. This skewed age distribution indicates that most failure observations come from relatively young equipment, which may make failure patterns toward early-time issues rather than long-term use.

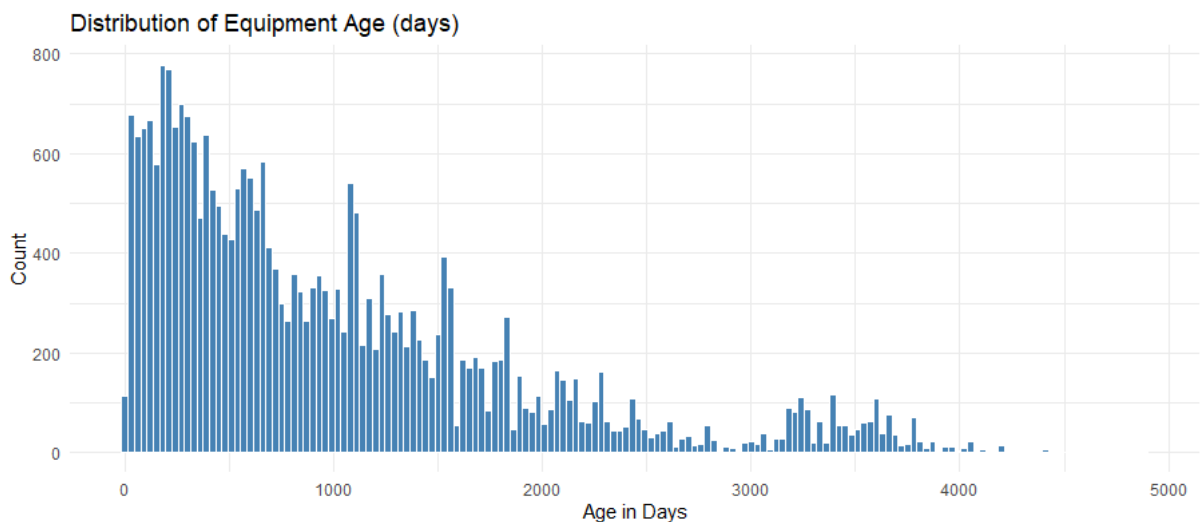


Figure 2.4: Distribution of equipment age(days)

Overall, the exploratory analysis suggests that wheel failures are rare, distribute unevenly across failure types, and associated with both mileage and equipment age. High flange and other failures are more common at higher mileage, while high impact and thin flange failures appear at lower ranges. The age distribution indicates limited long-term failure data, which is important to consider when modeling. These findings provide guidance for feature engineering in subsequent sections.

3 Modeling and evaluation

3.1 Feature Engineering and Selection

After data preprocessing and data enrichment on our dataset we ended up with 43 features describing wheel profiles, mileage, alerts, and vertical dynamics, to enhance the predictive power of the model and capture operational nuances of railcar behavior, we engineered a set of domain-informed features. These variables reflect temporal, mechanical, and performance-related aspects of equipment usage, failure history, and dynamic loading conditions. By integrating both raw sensor inputs and derived metrics, the feature set aims to improve interpretability and support robust failure prediction within a 30-day period.

By transforming raw attributes into derived indicators, we enhance the dataset with domain-informed representations that reflect cumulative stress, operational efficiency, and historical degradation. This improves predictive power by aligning the data with the physical processes underlying wheel wear and failure.

In our project, 13 features are engineered to complement and underpin the 43 raw attributes. These features are designed to capture aging effects, historical failure dependencies, and aggregated measures of stress and efficiency.

Feature engineering expands the predictive capacity of the dataset but together with the raw features introduces high dimensionality in the feature space, moreover not all features are equally relevant to the task and irrelevant features can lead to:

- Overfitting brought about by noise, where the model may memorize noise instead of learning generalizable patterns.
- Multicollinearity where redundant features can distort importance scores and weaken model interpretability.
- Computational inefficiency, where training times increase significantly with unnecessary features.

We employed both filter-based and wrapper-based feature selection approaches to balance computational efficiency with task-specific optimization.

3.1.1 Feature Engineering

Engineered Features

1. **Equipment Age in days**
it represents the elapsed service time of a wheel since its last application.
2. **Previous Failure History**
Encodes whether the equipment has failed previously.
3. **Cumulative Mileage** Represents the total distance traveled by a wheelset since installation.
4. **Total Travel Miles** Measures total operational exposure, accounting for both loaded and unloaded conditions.
5. **Utilization Ratio** Reflects the intensity of loaded service of the wheel relative to overall travel.
6. **Travel Efficiency** this ratio captures how efficiently the wheel accumulates mileage over its service life.
7. **Average Speed** Provides an approximation of average operational speed, useful when direct speed measurements are missing.
8. **Dynamic Loading Score** this is a composite metric combining dynamic vertical, dynamic ratio, hunting index, left-right imbalance, and percent load. it aggregates multiple dynamic stress indicators into a unified measure of load irregularities and vibration-induced degradation.
9. **Shock Stress** Quantifies peak stress events from maximum vertical relative to baseline vertical loading.
10. **Stress Ratio** This ratio expresses proportional stress fluctuations between the vertical force and the baseline.
11. **Applied Year and Month** they both captures temporal seasonality and maintenance practices linked to calendar cycles.
12. **Applied Age** Measures the elapsed time (in days) between the date of wheel application and the reference record month, capturing longer-term degradation trajectories aligned with monthly maintenance logs.

3.1.2 Feature selection

With 56 engineered and raw features in this study, feature selection was employed to identify subsets of features that maximize predictive performance while minimizing dimensionality. This study adopted a multi-strategy approach, employing filter, wrapper, and recursive elimination techniques to systematically evaluate feature subsets.

Filter-Based Feature Selection Filter methods assess the relevance of individual features independently of the final predictive model. Importance filter was applied:

- Random Forest Importance, which is Impurity based and quantifying the contribution of features to decision splits within a tree ensemble. This approach identified max vertical and shock stress among the most predictive.

model-based importance highlighted nonlinear interactions. Highlighting ideal number of top features to be around 15 without overfitting.

Wrapper-Based Feature Selection Wrapper methods integrate feature selection into the model training process, directly optimizing predictive performance. Two strategies were applied:

- Random Search Wrappers: Randomly sampled feature subsets were evaluated with Random Forest classifiers using log-loss as the objective.
- Genetic Algorithm Search: Simulated evolutionary selection of feature subsets, incorporating crossover, mutation, and elitism.

Both wrappers identified similar optimal subsets, with strong emphasis on features derived from mileage, stress imbalance, and dynamic vertical load metrics. Wrapper-based methods consistently outperformed filter-only subsets in cross-validation, underscoring their ability to account for feature interactions.

Comparative Analysis Filter vs. Wrapper: Filter methods were computationally cheap but less adaptive, while wrappers improved predictive accuracy by considering feature interactions. Both wrappers performed better than filter importance reinforcing the importance of adaptability of the features to the model in predictive maintenance as visualized in Figure 3.1.

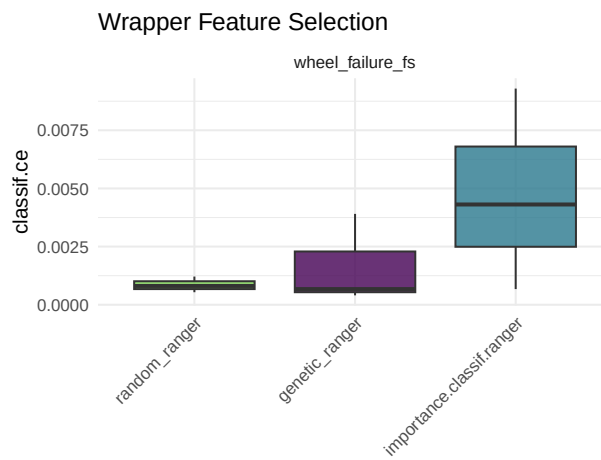


Figure 3.1: wrappers vs importance filter

Final Feature Set and Justification

The final selected subset of 14 features combined the strengths of wrapper-based selection using random search, balancing predictive performance and interpretability. Notably: High-value engineered features retained Dynamic Loading Score, left-right imbalance, hunting index, total travel miles.

Low-value or redundant features dropped: Index-like variables, duplicated site IDs, and weakly correlated categorical encodings.

Table 3.1: Selected Features

applied_age_days	applied_year	average_speed	dynamicvertical
flangeangle	flangethickness	huntingindex	leftrightimbalance
leveltwoalert	maxvertical	rimthickness	siteid
total_travel_miles	trainspeed.y		

3.2 Benchmarking and Model Selection

Model benchmarking to ensure that predictive performance is evaluated systematically across multiple algorithms under consistent experimental conditions to ensure consistency and incremental improvement in the predictive ability of our model is critical in our predictive maintenance projects. In the context of railway wheel failure prediction, benchmarking allows us to

assess trade-offs between interpretability, robustness to imbalance, and predictive accuracy.

This chapter presents a benchmarking study of five models — Random Forest (Ranger), XG-Boost, Logistic Regression, Support Vector Machines (SVM), and a Featureless Baseline — evaluated on the Feature engineered railway wheel dataset and subsequently on the selected features. We incorporated preprocessing (imputation, encoding, balancing) and dimensionality reduction Principal Component Analysis (PCA) pipelines to evaluate and achieve best results.

3.2.1 Benchmarking full model

Data Partitioning: The dataset was split into training (80%) and testing (20%) subsets. Date columns (record month, applied date, train date) are excluded from modeling after feature engineering, as they functioned only as temporal indices. This ensured that all models were trained and evaluated on the same feature space.

Task Creation: A multiclass classification task in Figure 3.2 was defined with **failurereason** as the target and metrics log-loss computed with clear reference to the failure events.

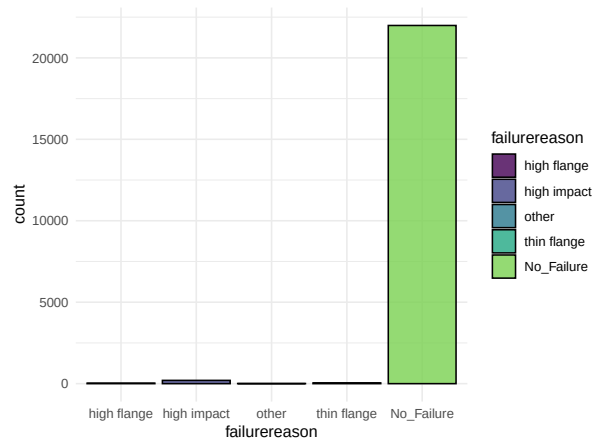


Figure 3.2: Task with failure reason as target

Out task reveals highly imbalanced classes with **No failure** being at 98.75% and other failure events being significantly rare. This skewed distribution poses challenges for both model training and evaluation. we will therefore consider undersampling strategy when it comes to our **No Failure** class and apply Logarithmic Loss (LogLoss) as our primary evaluation metric. LogLoss penalizes confident but incorrect predictions more heavily than uncertain ones, making it particularly suitable for probabilistic classifiers in high-stakes scenarios like predictive maintenance.

Resampling Strategy: 5-Fold cross-validation (CV) was employed, balancing computational feasibility with robustness. CV ensured that results were not biased by a single train-test split.

Preprocessing Pipelines: Preprocessing was structured as a graph to ensure interpretability, reproducibility, and minimal distortion of operational patterns. The workflow began with the removal of constant features and dealt with factor-type features using the one-hot encoding method. The method of undersampling aims to ensure that we do not discard vital operational patterns from the no-failure class, and scaling is applied where required, for example, in SVM and logistic regression. To address the extreme class imbalance, where the "No Failure" class accounts for 98.75% of observations, a targeted oversampling strategy was applied with a 4:1

ratio to address the imbalance issue.

Learners: Four learners and a baseline featureless learner are benchmarked with preprocessing graphs embedded:

- **Random Forest (Ranger)** which robust to non-linearities and feature interactions.
- **XGBoost:** Gradient-boosted trees with hyperparameters tuned for imbalance
- **Logistic Regression** - Linear baseline, interpretable and regularized to prevent overfitting and
- **SVM** with a Radial Kernel to captures non-linear decision boundaries and is sensitive to feature scaling

Evaluation Metrics Figure 3.3: The primary metric was log-loss, chosen for its ability to penalize miscalibrated probabilities in imbalanced settings.

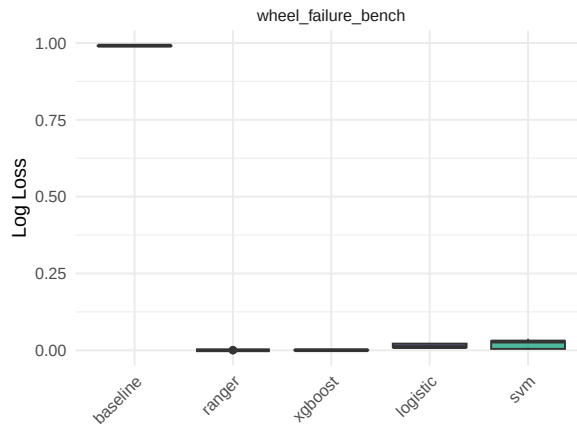


Figure 3.3: Full model Benchmark

Results the Benchmark Performance reveals that xgboost and Random forest achieved lowest logloss indicating superior probability calibration. Logistic Regression underperformed relative to ensemble methods, reflecting its limitations in modeling nonlinear relationships, and SVM showed moderate performance.

Model Training and prediction The results from Table 3.2 show the prediction based on logloss metric from the trained models show that XGBoost and Logistic Regression performed best.

Table 3.2: Multiclass Log Loss Scores for Different Learners

Learner	Log Loss
Random Forest	0.4480
XGBoost	0.0527
SVM	0.1201
Logistic Regression	0.0809

3.2.2 Principal Component Analysis

PCA with a rank of 20 was tested to evaluate whether dimensionality reduction improved model performance in Figure 3.4. Models with PCA performed significantly worse than models without, despite PCA ability to handle natively high-dimensional feature spaces, leading to the conclusion that PCA was universally not beneficial.

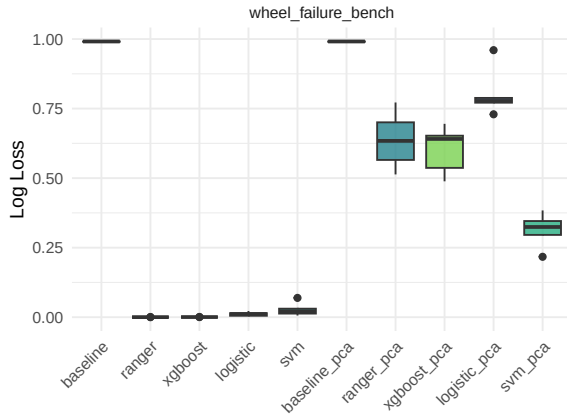


Figure 3.4: PCA vs Benchmark models

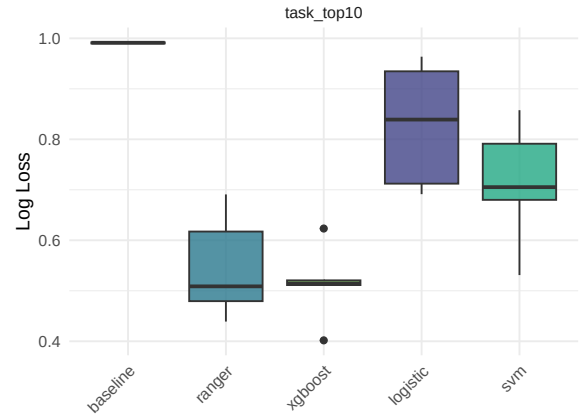


Figure 3.5: Benchmark on reduced features

XGBoost and Random forest still provided the best trade-off between predictive accuracy and robustness.

3.2.3 Benchmarking Model on Reduced Features

In the benchmarking a five-fold cross-validation strategy was applied to ensure robust performance estimation. The models compared were:

- **Baseline featureless classifier** (random guessing benchmark),
- **Random Forest (Ranger)** with tuned hyperparameters,
- **XGBoost**, a gradient-boosting framework optimized for speed and accuracy,
- **Logistic Regression**,
- **Support Vector Machines (SVM)** with radial kernel.

Log-loss was chosen as the main evaluation criterion, as it penalizes incorrect confident predictions and is suitable for multi-class classification.

Benchmarking Performance Figure 3.5 visualizes the log-loss scores from cross-validation. As expected, the baseline performed poorly, while Random Forest and XGBoost emerged as competitive models.

Prediction on Test Data Table 3.3 shows the log-loss on the unseen test dataset. Random Forest achieved the best generalization performance, followed by SVM and XGBoost.

Table 3.3: Test set log-loss results for trained models.

Model	Test Log-Loss
Random Forest (Ranger)	1.114
XGBoost	1.507
SVM	1.309
Logistic Regression	1.924

3.2.4 Model Selection

The results highlight that tree-based ensembles, particularly Random Forest, excel in handling heterogeneous engineered features and class imbalance present in railway maintenance data. Although XGBoost is often considered superior in tabular learning, its performance lagged behind Random Forest in this case. This could be attributed to limited feature interactions or the need for further hyperparameter optimization. Logistic regression underperformed, reflecting its limited capacity to capture nonlinear relationships in high-dimensional operational data. This study demonstrates that reducing the feature space to 14 informative variables enabled efficient benchmarking of multiple learners for predictive maintenance. Random Forest provided the most stable and lowest log-loss on both cross-validation and test evaluation, but Xgboost is more often superior upon hyperparameter Optimization. Future work will involve extended hyperparameter optimization of **XGBoost** to further improve predictive accuracy.

3.3 Hyperparamter Optimization

The hyperparameter optimization was performed on XGBoost where random search was applied across a parameter space including the number of boosting rounds (nrounds), learning rate (eta), tree depth (max_depth), and sampling ratios. Bayesian Optimization (via model-based optimization, MBO) was used in tuning. Table 3.4 summarizes the search space explored.

Table 3.4: Hyperparameter search space for XGBoost

Model	Parameter	Range
XGBoost	nrounds	50–300
	eta	0.01–0.3
	max_depth	3–10
	subsample	0.7–1.0
	colsample_bytree	0.7–1.0
	scale_pos_weight	10–30

A 3-fold outer cross-validation loop was applied to benchmark the tuned learners, while inner resampling with 3-fold cross-validation guided the optimization process. The primary metric for optimization was log-loss to capture trade-offs between predictive confidence and class sensitivity.

Results: Table 3.5 presents the predictive performance of the tuned XGBoost model.

Table 3.5: Predictive Performance of XGBoost model

Learner	Log-Loss
XGBoost (Tuned)	1.31

The results indicate that hyperparameter optimization improved XGboost model. XGBoost showed competitive performance, with a modest improvement in log-loss relative to its untuned benchmark.

The optimization results suggest that the marginal improvement of XGBoost indicates potential for further gains with more exhaustive hyperparameter tuning although in practice, the choice between models may also be influenced by computational constraints, interpretability, and deployment considerations.

Table 3.6: Confusion matrix of the tuned XGBoost model on the test set.

Truth / Response	High Flange	High Impact	Other	Thin Flange	No Failure
High Flange	10	0	0	0	398
High Impact	0	44	0	0	1273
Other	0	1	0	0	8
Thin Flange	0	0	0	7	987
No Failure	0	0	0	1	2840

Conclusion: Hyperparameter optimization confirmed Random Forest as the best-performing model for predicting wheel failures in the railway maintenance dataset. While XGBoost remains a competitive alternative, Random Forest provides superior performance with greater robustness, making it a suitable candidate for operational deployment. Future work will involve scaling hyperparameter optimization to larger search spaces and investigating model stacking or blending strategies.

3.4 Interpretability

In this section, we applied a suite of interpretable ML techniques to examine both the global structure and local decision logic of the classification model. These methods provide a deeper understanding of feature influence (via Feature Importance), individual prediction logic (via LIME-style and Shapley), with the number of samples capped at 50 per iteration to reduce feature dominance and ensure interpretive stability.

3.4.1 Feature Importance (Cross-Entropy Loss)

Using cross-entropy as the evaluation metric, we quantified the relative contribution of each feature to the predictive performance. The most influential variables are average speed, total travel miles, flange thickness, and dynamic vertical loading. Temporal features, such as the applied year and applied age, are also significant, suggesting a latent relationship between time and mileage of usage. Figure 3.6 displays 12 features with substantial importance, consistent with our assumption of the feature selection step.

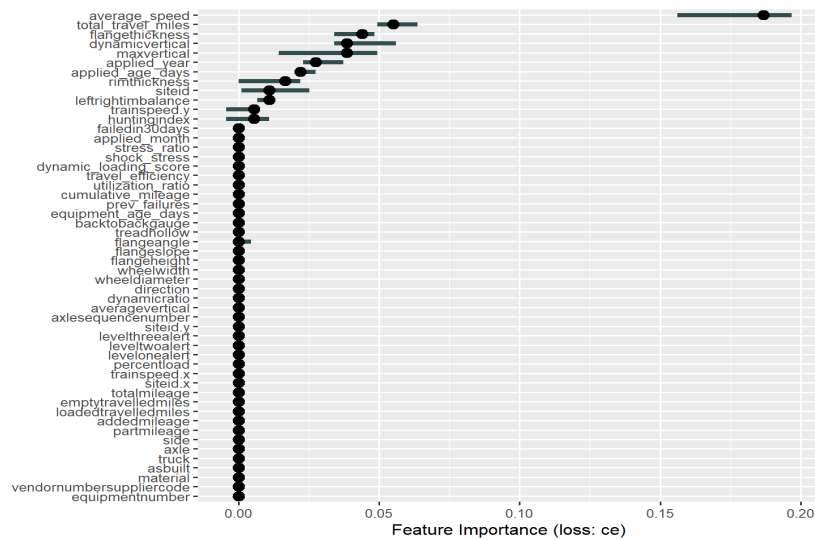


Figure 3.6: Feature importance by cross-entropy loss

3.4.2 Local Explanation (LIME-style)

Figure 3.7 illustrates the directional impact of three key features—failed in 30 days, vendor number (supplier code), and max vertical—across five structural categories. The binary feature, failed in 30 days, does not contribute to the impact for all five categories. Avoiding the result-related feature data leakage. The vendor number also contributes to the model's impact, which may reflect the quality of the parts supplied by the vendors. Furthermore, the maximum vertical has a significant effect on the flange-related failure.

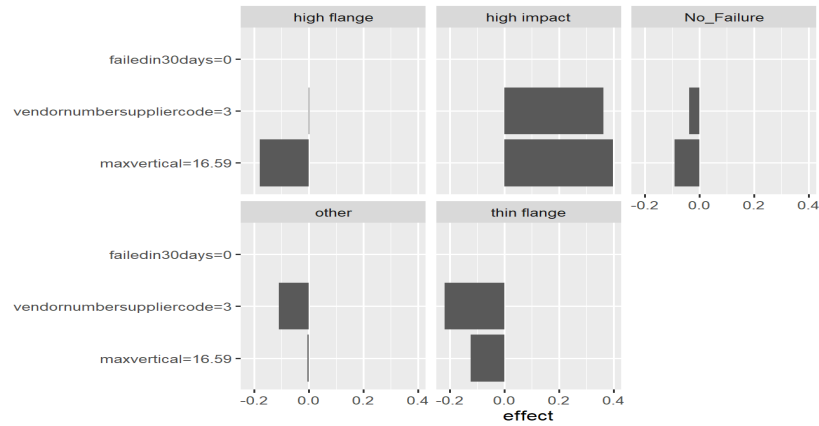


Figure 3.7: Result of Local Explanation

3.4.3 Shapley Value Analysis (SHAP)

To enhance model interpretability, we applied SHAP (SHapley Additive exPlanations) to decompose feature contributions across different failure modes. Rather than visualizing the full feature set, we selected the top five most influential features per class to foreground the structural logic behind each prediction. Across multiple failure categories—including high impact, thin flange, and no failure—features such as total travel miles, average speed, and applied age days, present a reliance on cumulative usage and temporal exposure as dominant drivers. Structural attributes, such as flange thickness, rim thickness, and maximum vertical, played class-specific roles. The “other” and “thin flange” categories exhibited more dispersed SHAP distributions, suggesting either lower signal strength or narrative dilution in these two categories.

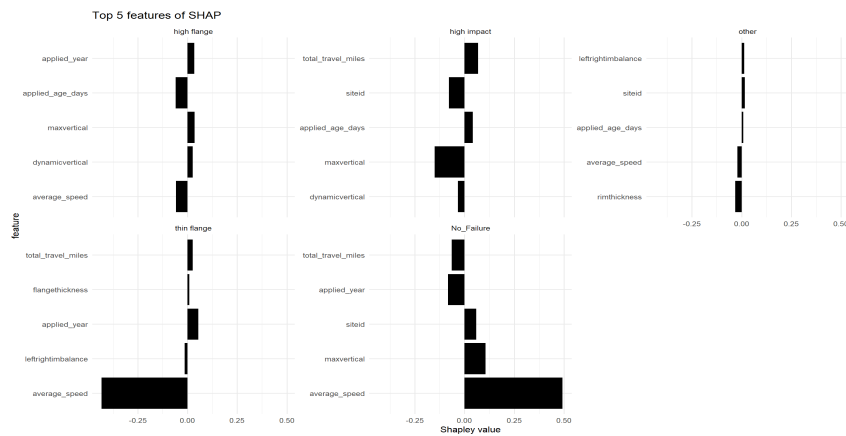


Figure 3.8: Result of Game-theoretic fairness

4 Conclusion

This project focuses on predictive analysis of train wheel failures using heterogeneous data collected from multiple system folders. We began by performing data cleaning and sampling, followed by merging datasets through common features to construct a table. To enhance the model's sensitivity to latent risk factors, we performed feature engineering—particularly on those reflecting physical wear and operational stress—that may not be captured in the raw datasets.

Given that the initial dataset contained over 50 features, we applied wrapper-based feature selection using random search to identify the most influential variables, which are no more than 15. This step was critical in reducing model complexity and mitigating the risk of overfitting. We then applied Hyperparameter Optimization (HPO) and AutoML tools to explore optimal model architectures and parameter configurations. Both results above show that XGBoost is consistently the most stable and deployable model for this task.

The analysis after training revealed that the most predictive features were closely tied to time and distance metrics, reinforcing the intuitive notion that cumulative usage and travel distance contribute directly to wheel failure prediction. In the feature importance rankings derived from cross-entropy loss, average speed held the highest weight, while load-related variables also showed strong associations with failure outcomes. These findings align well with physical expectations, lending interpretability and credibility to the model's behavior.

One of the major challenges in this task was severe class imbalance, which impaired the model's ability to learn from minority failure cases. We tried various strategies to construct our model and found that binary classification (failure or not) yields the most reliable results. In contrast, models trained solely on multi-class failure types lacked viability. The final model combined binary and multi-class features, but it must carefully control potential label leakage to ensure that binary features do not interfere with multi-class predictions. Our global interpretability results confirmed that there was no leakage, as the binary feature had zero impact on the final model, supporting the integrity of our modelling.

For data splitting, we initially adopted a proportional split, which carries the risk of data leakage between training and test sets. In later stages, we attempted temporal splitting for the multi-class task; however, poor model performance limited our ability to evaluate its effectiveness. Future work should introduce temporal splits into the current best-performing model and redesign sampling strategies, with a primary focus on increasing the samples of minority failure cases. This would enhance the model's learning and solve the issues of class imbalance.

Contributions of the Group Members

Table 4.1: Contributions of the group members

Section	Contributor
Introduction	Wenjing Yu
Data and preprocessing	Michael Kiboi & Shih-Hsien Hsu
Exploratory Data Analysis	Wenjing Yu
Feature Engineering & selection	Michael Kiboi
Modelling and Evaluation	Michael Kiboi & Shih-Hsien Hsu
Interpretability	Shih-Hsien Hsu
Conclusion	Shih-Hsien Hsu

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Dresden, 29th August 2025

Michael Mutahi Kiboi



Shih-Hsien Hsu



Wenjing Yu

